



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SIVASAKTHI & CO.

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

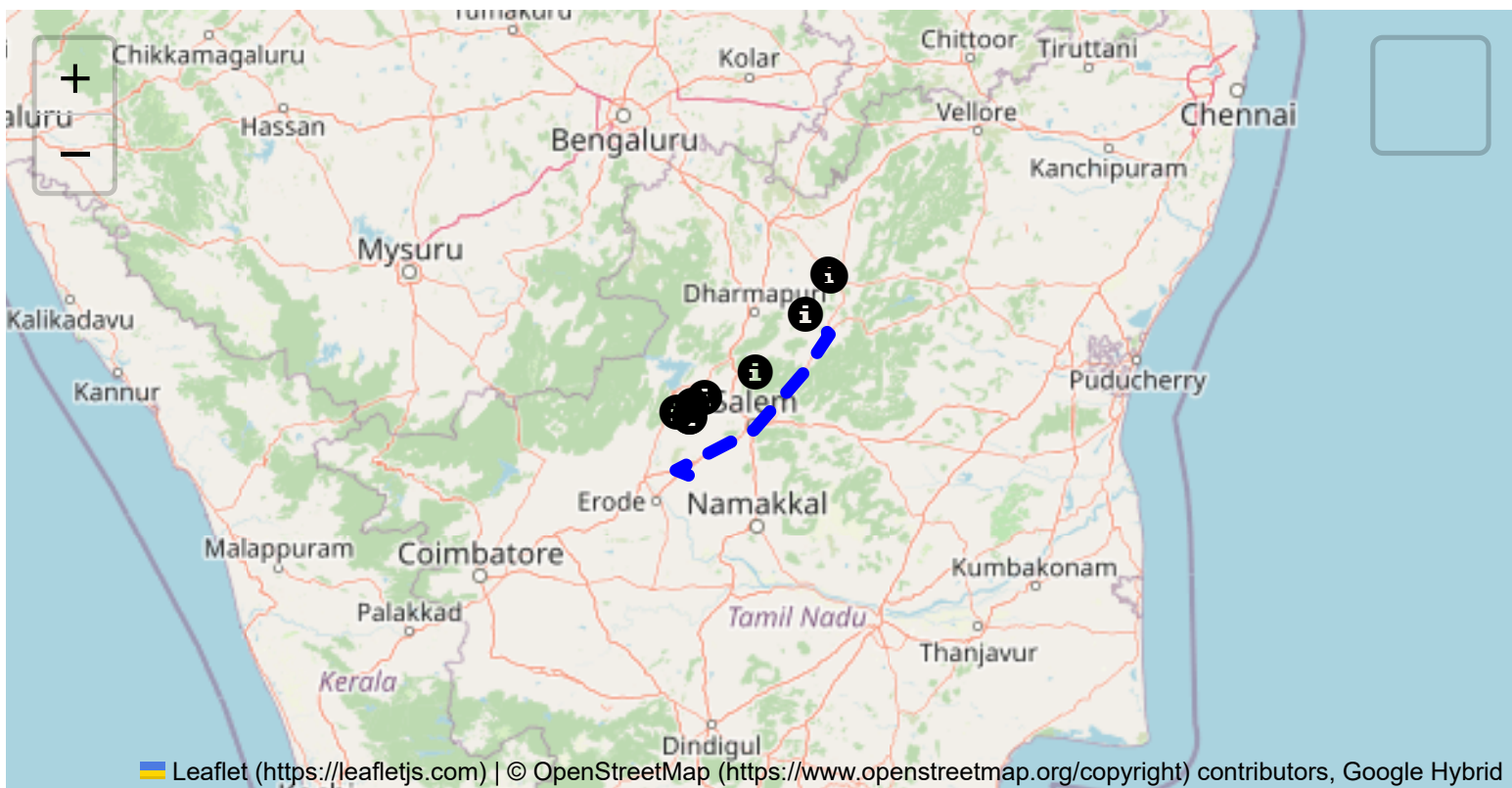
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

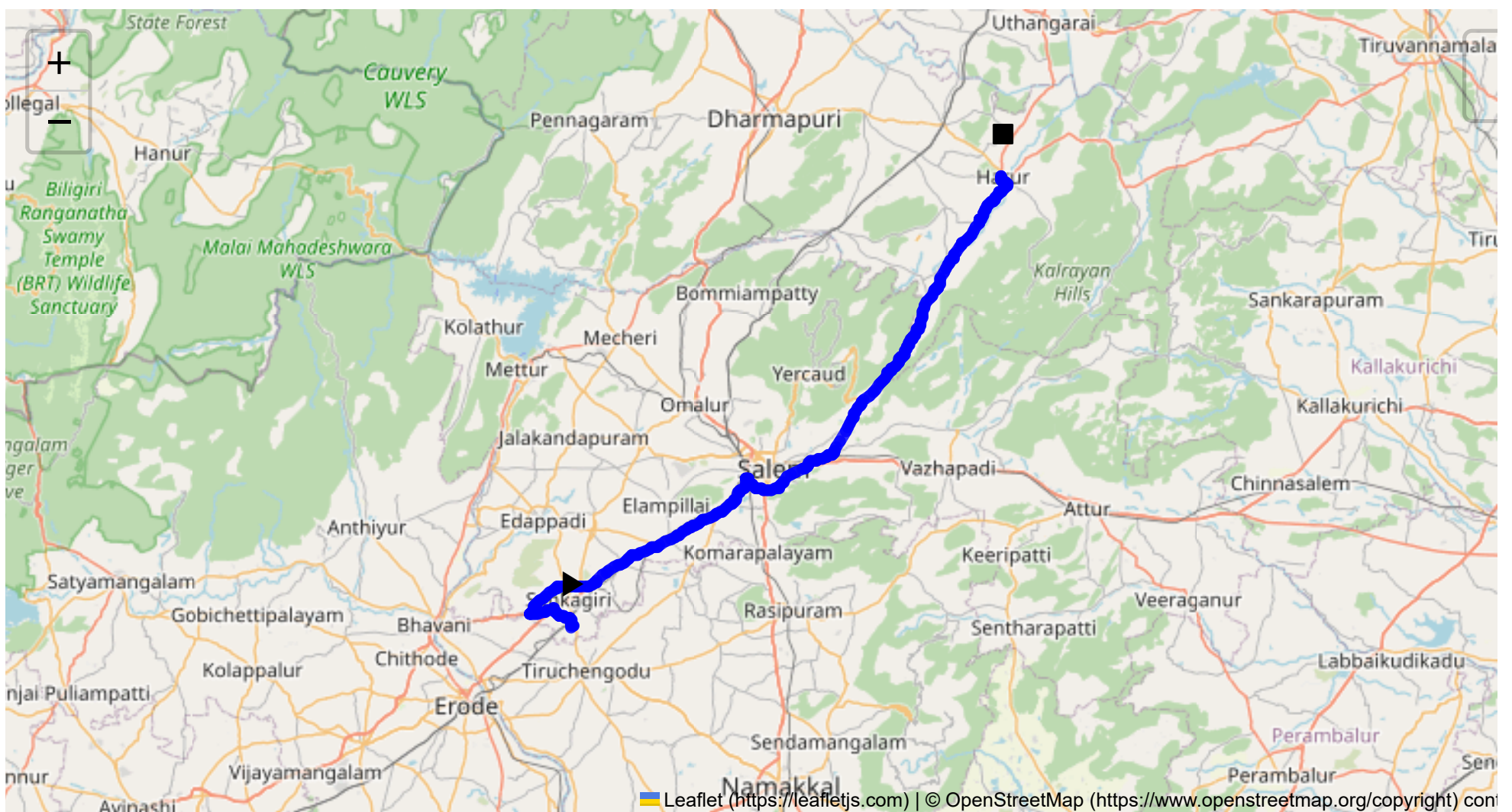
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 117.84 km
Estimated Duration: 2.8 hours
Adjusted Duration (Heavy Vehicle): 3.5 hours
Start: (11.4381, 77.8734)
End: (12.052225, 78.475166)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The described route winds through several significant regions in Tamil Nadu, starting from Sangagiri and ending in Periyar Nagar, Harur. The journey covers diverse landscapes, transitioning from rural stretches to more urban environments as it passes through Salem. Major waypoints in this route include Puthur, Padaiveedu, Vaikundam, Seelanaickenpatti, K R Nagar, and NH 179A, before concluding in Harur.

2. Typical Weather Conditions and Potential Hazards

- **Climate:** Tamil Nadu generally experiences a tropical climate, with hot summers, moderate monsoons, and mild winters.
- **Potential Hazards:** During the monsoon season (June to September), heavy rainfall can cause waterlogging and reduce visibility, increasing the risk of accidents. The summer months (March to June) can be extremely hot, affecting vehicle performance and driver endurance.

3. Traffic Patterns and Congestion

- **Peak Hours:** Typically, traffic congestion peaks in the morning (8 AM to 10 AM) and evening hours (5 PM to 8 PM), especially around Salem city.
- **Congestion Areas:** The Salem city area, particularly near Seelanaickenpatti, is prone to heavy traffic. Bhavani Main Road and Salem By-pass experience regular congestion due to their roles as major thoroughfares.

4. Road Quality and Infrastructure

- **Overall Quality:** Generally, the main highways are well-maintained. However, certain rural sections might have narrow lanes and need better maintenance.
- **Infrastructure:** Be cautious around ongoing construction zones, which may change regularly.

5. Alternative Routes for Emergencies

- **Primary Alternative:** In case of severe congestion or obstacles, using the state highways SH-79 and SH-20 may provide alternate paths. These routes are typically less direct but can help avoid problematic sections.

6. Local Regulations for Hazardous Material Transport

- **Regulations:** Transportation of hazardous materials requires adherence to local and national safety guidelines, including securing proper permits and ensuring vehicle compliance with hazardous material transport norms.

7. Historical Incidents

- **Previous Incidents:** Review data indicates occasional road accidents involving heavy vehicles, often attributed to road hazards during monsoon and the congested Salem area. It underlines the need for cautious driving, especially in such conditions.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Zones:** The route passes through semi-urban areas with residential communities. It is crucial to minimize noise and pollution. Avoid idling in populated areas and reduce speed near schools and hospitals.

9. Communication Coverage

- **Network Availability:** Urban sections such as Salem have robust network coverage. However, certain rural stretches might experience sporadic service gaps. Depend on offline maps and consider carrying a satellite phone for emergencies in dead zones.

10. Emergency Response Times

- **Response Times:**
 - **Urban Sections (e.g., Salem):** Typically between 15 to 30 minutes.
 - **Rural Sections:** Response times may extend to 45 minutes to an hour due to limited accessibility.

12. Overall Summary of Risk Assessment

The analyzed route, while feasible for the transport of hazardous materials, presents several challenges including monsoon-related hazards and urban traffic congestion. Adequate preparation, such as monitoring weather forecasts and peak traffic times, adherence to local regulations, and readiness for communication challenges, is necessary to mitigate risks. Emergency protocols should be established for each segment to improve responsiveness and ensure the safety of the transport. Regular monitoring for road conditions and updated advisories from local authorities will enhance route safety further.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	U-Turn	High	12.0543064, 78.4762765	10 KM/Hr
2	Turn	High	11.43961, 77.87341	15 KM/Hr
3	Turn	High	11.44024, 77.87528	15 KM/Hr
4	Turn	High	11.44898, 77.87410	15 KM/Hr
5	Turn	High	11.45352, 77.85698	15 KM/Hr
6	Turn	Medium	11.45847, 77.85140	30 KM/Hr
7	Turn	Medium	11.45898, 77.85135	30 KM/Hr
8	Turn	High	11.46303, 77.84945	15 KM/Hr
9	Turn	Medium	11.45543, 77.81740	30 KM/Hr
10	Turn	High	11.45631, 77.81668	15 KM/Hr
11	Turn	Medium	11.53552, 77.95763	30 KM/Hr
12	Turn	Medium	11.64158, 78.11808	30 KM/Hr
13	Turn	Medium	11.64210, 78.11860	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
14	Turn	High	11.64172, 78.12154	15 KM/Hr
15	Turn	Medium	11.64098, 78.12163	30 KM/Hr
16	Turn	High	11.63972, 78.12083	15 KM/Hr
17	Turn	Medium	11.62826, 78.13337	30 KM/Hr
18	Turn	High	11.66372, 78.20649	15 KM/Hr
19	Turn	Medium	11.67124, 78.22961	30 KM/Hr
20	Turn	High	11.67111, 78.23134	15 KM/Hr
21	Turn	High	11.67187, 78.23142	15 KM/Hr
22	Turn	Medium	11.67427, 78.23892	30 KM/Hr
23	Turn	Medium	11.68018, 78.24251	30 KM/Hr
24	Turn	Medium	11.81208, 78.34071	30 KM/Hr
25	Turn	Medium	11.83518, 78.35305	30 KM/Hr
26	Turn	Medium	11.99554, 78.44553	30 KM/Hr
27	Turn	Medium	12.04286, 78.48352	30 KM/Hr
28	Turn	Medium	12.04440, 78.48380	30 KM/Hr
29	Turn	High	12.04973, 78.47494	15 KM/Hr
30	Turn	Medium	12.05033, 78.47468	30 KM/Hr
31	Blind Spot	Blind Spot	12.05427, 78.47634	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
4	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
5	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
8	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
12	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
14	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
15	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
16	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
17	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
18	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
19	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
20	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
21	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
22	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
23	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
26	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
27	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium
30	hospital	Government Hospital Chinnankuppam	12.0110294, 78.4553224	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
6	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
9	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
13	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
24	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
25	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium
28	school	Indian Matriculation Higher secondary school	12.006299, 78.4531204	30 km/h	Medium
29	school	Indian Matriculation Higher Secondary school	12.0061755, 78.4533388	30 km/h	Medium
31	school	Kurinji Matriculation Higher Secondary School	12.0156853, 78.4579056	30 km/h	Medium
32	school	Chinnan Kuppam Government High School	12.0190729, 78.4581955	30 km/h	Medium
33	college	Annai Arts and Science College	12.0217961, 78.462795	30 km/h	Medium
34	school	Jayam Vidhyalaya Matriculation Higher Secondary School	12.0264103, 78.4656553	30 km/h	Medium
35	school	Harur International School (CBSE)	12.0299967, 78.4693492	30 km/h	Medium
36	school	Government Girls Hr Sec School	12.0488709, 78.4762447	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 12.0543064, 78.4762765



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44024, 77.87528



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45898, 77.85135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45543, 77.81740



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.53552, 77.95763



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64158, 78.11808



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64210, 78.11860



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64172, 78.12154



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64098, 78.12163



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63972, 78.12083



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.62826, 78.13337



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66372, 78.20649



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67124, 78.22961



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67111, 78.23134



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67187, 78.23142



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67427, 78.23892



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68018, 78.24251



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.81208, 78.34071



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.83518, 78.35305

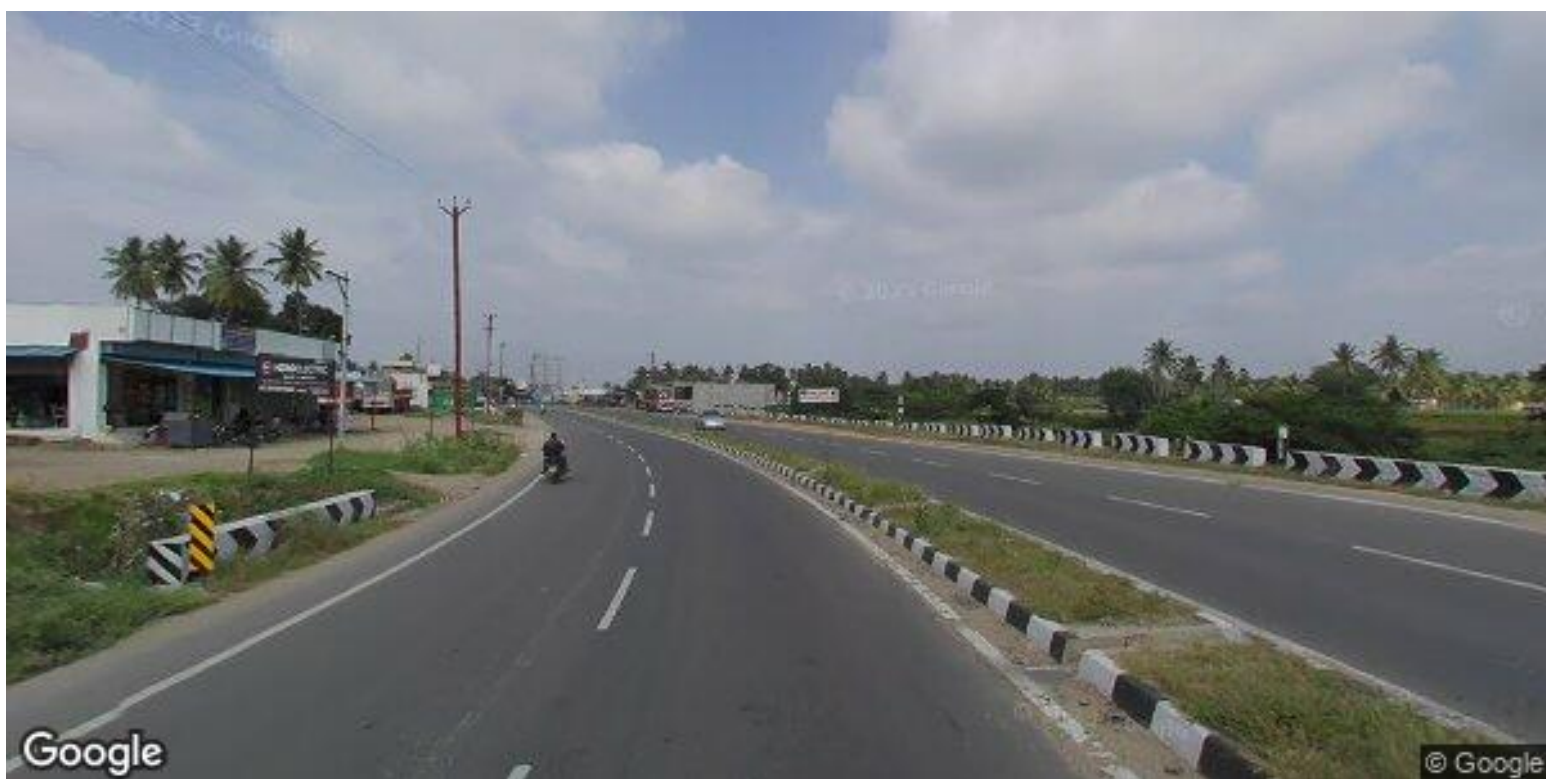


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.99554, 78.44553



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.04286, 78.48352



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.04440, 78.48380



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.04973, 78.47494



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.05033, 78.47468



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.05427, 78.47634
